EEG Oscillatory Dynamics During Increasing Working-Memory Demand: An Analysis of the N-Back Task

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Abstract

This study investigated how EEG oscillatory activity varied across four levels of working-memory demand in an N-back task. EEG recordings from 18 participants were processed to quantify theta, alpha, and beta power and to examine condition-related differences. Alpha activity showed the clearest condition-related modulation, with the strongest pairwise difference between the 2-back and 4-back conditions. Theta activity did not show a significant overall condition effect, and the beta effect did not survive correction for multiple comparisons. The exploratory association between alpha change and recorded behavioral accuracy was not statistically significant. A post-hoc audit of the publicly available task implementation raised a concern about the validity of the recorded *Matched* labels used to derive behavioral accuracy; because the exact software version used during acquisition could not be independently verified and the event markers do not contain the stimulus letters, behavioral results are interpreted cautiously. The primary conclusions therefore concern the EEG spectral analyses.

Keywords: EEG, working memory, N-back, spectral analysis, theta, alpha, beta, independent component analysis

1. Introduction

Working memory supports the temporary maintenance and updating of information during ongoing cognition. The N-back task operationalizes working-memory demand by asking participants to compare each incoming stimulus with the stimulus presented a specified number of positions earlier [1]. Increasing the value of N changes the number of intervening items that must be maintained and updated.

Electroencephalography (EEG) records electrical potential differences at the scalp and provides high temporal resolution for studying neural activity. Scalp EEG is largely associated with summed, locally synchronous postsynaptic activity in cortical neuronal populations [2]. Neural oscillations are commonly studied in relation to cognitive processing, but their relationship with memory demand depends on task design, frequency range, and analysis interval. Prior work has discussed roles for theta activity in working-memory control and organization, and for alpha activity in attentional selection and inhibition [3–5]. However, the direction of a band-power

change should not be assumed in advance: reviews report variability in the direction of alpha effects across working-memory tasks [6].

This study examined EEG oscillatory activity across 1-back, 2-back, 3-back, and 4-back conditions. The primary research question was: *How does EEG oscillatory activity change with increasing working-memory demand, and are these neural changes related to behavioral performance?* The principal neural measures were theta, alpha, and beta band power. Behavioral analyses were intended as secondary and exploratory measures.

2. Background

2.1 EEG Frequency Bands and Working Memory

Three frequency bands were examined. Their predefined frequency ranges were:

$$\theta : 4\text{–}8\,\text{Hz},$$
$$\alpha : 8\text{–}13\,\text{Hz},$$
$$\beta : 13\text{–}30\,\text{Hz}.$$

Theta oscillations have been associated with control and organization of working-memory processing [4]. Alpha activity has been linked to attention, inhibition, and memory-related processes, although its direction of modulation varies across tasks and conditions [3, 5, 6]. Beta activity has also been investigated in relation to cognitive processing, but findings are less consistent across working-memory studies [6]. Accordingly, all three bands were examined empirically rather than assuming a particular direction of change.

2.2 Spectral Analysis

Power spectral density (PSD) estimates how signal power is distributed across frequencies. Welch's method estimates PSD by dividing a signal into overlapping segments, computing modified periodograms, and averaging them to obtain a more stable estimate [7]. Integrating the PSD within a predefined frequency range provides a band-power measure for comparisons across conditions.

3. Methodology

3.1 Experimental Design and Dataset

The study used the publicly available OpenNeuro dataset `ds007169`, titled *Multimodal Cognitive Workload n-back Task, 4 Difficulties* [12]. The dataset contained 20 participants; participants `sub-013` and `sub-017` were excluded based on the dataset metadata, leaving 18 participants. EEG was recorded from 19 scalp electrodes at 250 Hz. The task contained four conditions (1-back through 4-back), with 100 experimental trials per condition. Tutorial trials were excluded. The dataset is organized using the EEG-BIDS conventions for representing electrophysiology data and metadata [2].

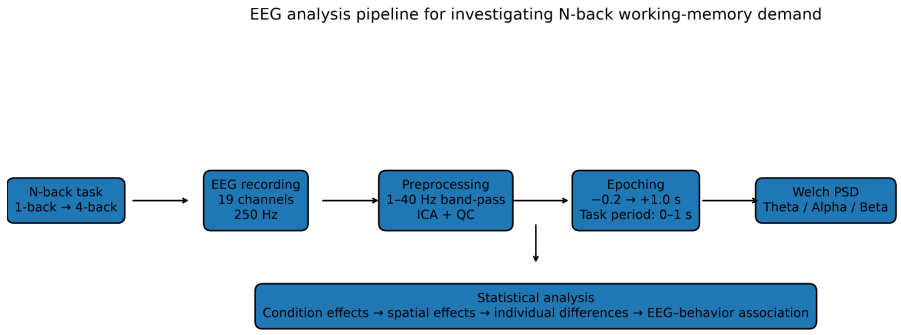


Figure 1. Overview of the EEG analysis pipeline used in this study.

3.2 EEG Preprocessing

Raw EEG recordings were provided in BrainVision format and loaded using MNE-Python, a software package for EEG/MEG processing and analysis [8]. The recordings contained 19 scalp EEG channels, one ECG channel, and four miscellaneous channels. The analysis focused on the 19 EEG channels: Fp1, Fp2, F3, F4, F7, F8, Fz, C3, C4, Cz, T3, T4, T5, T6, P3, P4, Pz, O1, and O2. A signal-scaling correction was applied before subsequent processing. A standard 10–20 montage was assigned to support scalp-topography visualization; these were approximate standard electrode positions rather than individually digitized coordinates.

The continuous EEG was band-pass filtered between 1 and 40 Hz using MNE-Python. This range retained the frequency bands of interest while attenuating slow fluctuations and higher-frequency activity outside the planned analysis range.

3.2.1 ICA-Based Artifact Investigation

Independent Component Analysis (ICA) was applied to the 19 EEG channels using the FastICA implementation in MNE-Python. The decomposition used 19 components, random seed 42, and automatic iteration settings. ICA was fitted using every third sample to reduce computational cost; this decimation applied during component estimation did not change the original 250 Hz

sampling rate. ECG and miscellaneous channels were excluded from the decomposition. ICA is widely used to separate EEG mixtures into components that can be inspected for artifact contributions [9, 10].

Because no dedicated electrooculography (EOG) channels were available, Fp1 and Fp2 were used as supporting indicators when investigating possible ocular artifacts. Candidate components were assessed using scalp topography, temporal waveform, frequency characteristics, and relationships with the frontal channels. No single indicator was treated as conclusive, because frontal EEG activity can also reflect neural sources. Component identification therefore remained interpretative.

For participant `sub-001`, component ICA000 showed a pronounced frontal distribution and large, slow temporal deflections around 98–103 seconds. Within the examined interval, it correlated with Fp1 ($r = 0.746$) and Fp2 ($r = 0.733$). Removing the component substantially reduced the large slow frontal deflections, although a broader disturbance remained. These observations supported the interpretation that ICA000 contributed to the frontal artifact, but did not establish that it accounted for all contamination.

3.2.2 ECG Sensitivity Analysis

The ECG recording was examined separately to investigate a possible relationship between cardiac activity and EEG alpha power. An exploratory trial-level Spearman analysis across participants yielded a median absolute correlation of 0.029 and a maximum absolute correlation of 0.177. Two participants showed nominally significant correlations, while constant-input cases occurred for several participants. ECG was not explicitly regressed out or removed from the final EEG preprocessing pipeline. This sensitivity analysis therefore does not establish the absence of cardiac contamination.

3.2.3 Epoching and Artifact Screening

Experimental event markers were used to identify N-back trials, and tutorial trials were excluded. Each trial was segmented from −0.2 to +1.0 s relative to stimulus onset. The pre-stimulus interval was designated as baseline, but only the 0–1 s post-stimulus interval was used for the primary spectral analysis; the short pre-stimulus interval was not used for baseline-normalized PSD estimation.

Epoch-level artifact screening used maximum and median peak-to-peak amplitudes across the 19 EEG channels. Participant-specific 95th-percentile thresholds were calculated for both measures. An epoch was flagged as a candidate artifact when both thresholds were exceeded, and candidate epochs were reviewed as part of quality control. Of 7,200 initial experimental epochs, 23 were rejected, leaving 7,177 epochs for spectral analysis (0.32% rejection).

3.3 Spectral Analysis

PSD was calculated for each retained epoch and each EEG channel using Welch's method [7]. The analysis used 125-sample segments (0.5 s) with 62-sample overlap and restricted the PSD to 1–40 Hz. Band power was obtained by integrating the PSD within the three predefined frequency bands, expressed in μV^2.

The resulting feature table contained 136,363 channel-level observations. Each row represented one retained epoch at one electrode and included participant, epoch, condition, electrode, and three band-power measurements.

To express condition-related power relative to each participant's 1-back condition, band power was transformed using the natural logarithmic ratio:

$$L_{i,c,b} = \ln\left(\frac{P_{i,c,b}}{P_{i,1,b}}\right), \tag{1}$$

where $P_{i,c,b}$ is the power for participant i, condition c, and frequency band b, and $P_{i,1,b}$ is the corresponding 1-back power. Thus, the 1-back value equals zero; positive and negative values indicate increases and decreases, respectively, relative to 1-back.

For the primary repeated-measures analysis, normalized values were averaged across the 19 EEG channels for each participant, frequency band, and condition, yielding 72 participant-condition observations (18 participants $\times$ 4 conditions) per band. An electrode-wise analysis also examined the alpha-power contrast between 2-back and 4-back.

3.4 Statistical Analysis

Analyses were conducted at the participant level, so participants rather than epochs or channels were the units of inference. Repeated-measures ANOVA was used to test condition effects, with the Friedman test as a non-parametric robustness analysis. Kendall's W was used to describe Friedman effect magnitude. Because three frequency bands were examined, the Bonferroni-adjusted significance threshold was $p = 0.0167$.

Pairwise condition comparisons used Wilcoxon signed-rank tests, with Bonferroni adjustment across the six comparisons. For the electrode-wise alpha analysis, paired Wilcoxon tests were performed at each of the 19 electrodes and p-values were adjusted using the Benjamini–Hochberg false-discovery-rate procedure [11]. An exploratory Spearman rank correlation examined the association between the change in alpha activity from 2-back to 4-back and the corresponding change in recorded behavioral accuracy. Leave-one-participant-out analyses tested whether the alpha contrast depended disproportionately on any single participant.

4. Results

4.1 Behavioral Performance: Interpretation Caveat

The initial analysis of the dataset's recorded behavioral labels yielded mean accuracies of 43.5%, 61.7%, 73.9%, and 69.6% for 1-back through 4-back, respectively. These values were reproduced by independently recalculating accuracy from the dataset's `Matched` and `KeyPress` fields. However, a post-hoc audit of the publicly available task implementation identified a potential indexing problem in the code used to calculate `Matched`: the expression `stepsArray[3-numberBack]` appears inconsistent with the position expected for a standard N-back comparison. The public repository history shows this expression in earlier versions as well, but the exact source version used during the recorded sessions could not be independently verified. The raw event markers do not contain the stimulus letters required to reconstruct trial-wise target status independently [12, 13].

For this reason, the recorded accuracy values and the behavioral comparison across conditions should be treated as provisional. The exploratory association between EEG alpha change and these recorded accuracy values is also interpreted cautiously. This limitation does not by itself invalidate the EEG spectral calculations, which were based on EEG epochs and their condition labels rather than on `Matched` values.

4.2 Theta Power

Theta activity did not show a significant condition effect. Mean theta log-ratio values were 0, −0.0395, −0.0054, and 0.0107 for 1-back through 4-back, respectively. The Friedman test was not significant ($\chi^2(3) = 0.60$, $p = 0.896$, Kendall's $W = 0.011$), indicating no reliable change in theta activity across the four conditions.

4.3 Alpha Power

Alpha activity showed the clearest condition-related modulation. Mean alpha log-ratio values were 0, −0.1124, −0.0642, and 0.0362 for 1-back through 4-back. The Friedman test indicated a significant condition effect ($\chi^2(3) = 17.67$, $p = 0.000515$, Kendall's $W = 0.327$), which remained significant after correction for the three frequency bands.

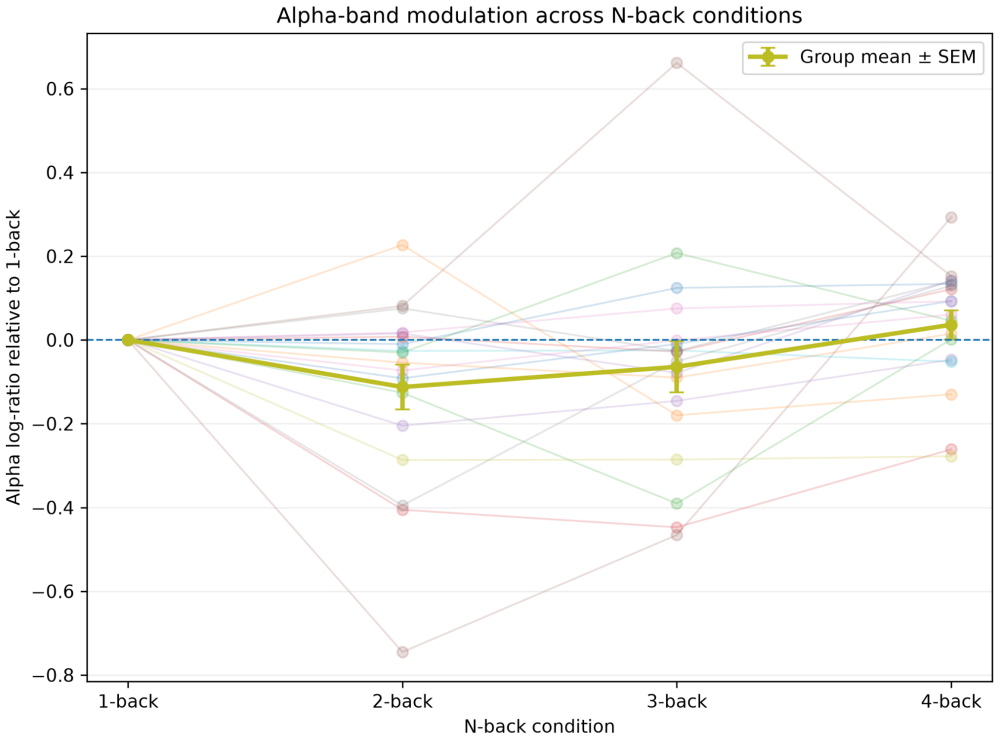


Figure 2. Participant-level alpha log-ratio values across N-back conditions. Values are relative to each participant's 1-back power.

4.4 Beta Power

Beta log-ratio means were 0, −0.0454, 0.0188, and 0.0513 for 1-back through 4-back. The Friedman test produced $\chi^2(3) = 8.67$, $p = 0.0341$, with Kendall's $W = 0.161$. This result did not remain significant after applying the corrected threshold of $p = 0.0167$.

4.5 Alpha: 2-Back Versus 4-Back

The alpha-power contrast between 2-back and 4-back was examined in greater detail. Alpha activity was significantly higher in the 4-back condition than in the 2-back condition (Wilcoxon signed-rank test: $W = 18$, unadjusted $p = 0.00193$, Bonferroni-adjusted $p = 0.01158$). The mean paired difference was 0.1486, the median difference was 0.1200, and the rank-biserial effect size was 0.7895.

A leave-one-participant-out sensitivity analysis showed that the contrast remained significant in all 18 analyses, with p-values ranging from approximately 0.000046 to 0.003845.

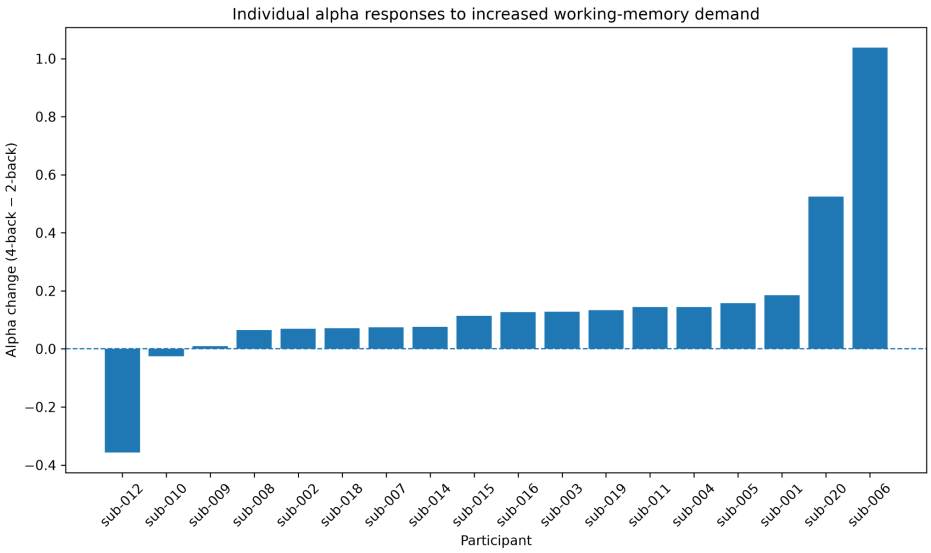


Figure 3. Individual alpha-power log-ratio differences (4-back minus 2-back).

4.6 Spatial Distribution of the Alpha Effect

The 2-back versus 4-back alpha difference was examined separately across the 19 electrodes. After Benjamini–Hochberg FDR correction, significant differences were observed at nine electrodes. Sixteen of the 18 participants showed a positive 4-back minus 2-back alpha difference, indicating that the group-level increase was not attributable to only a small subset of participants.

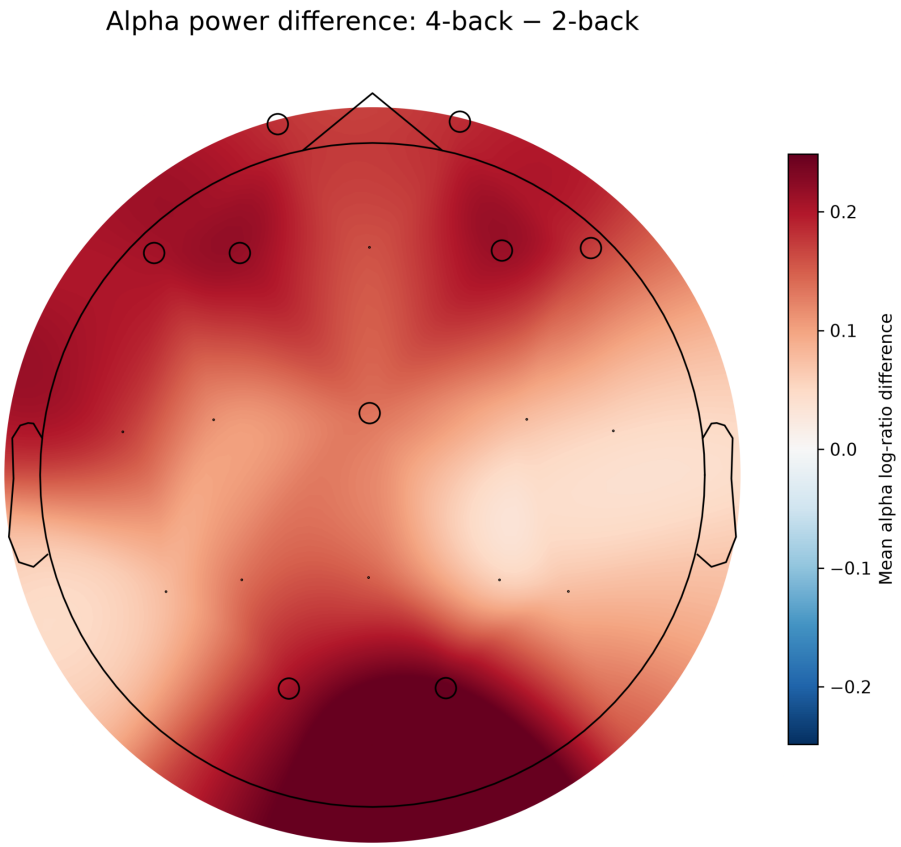


Figure 4. Scalp distribution of the mean alpha log-ratio difference (4-back minus 2-back). Electrode positions use an approximate standard montage and should not be interpreted as source localization.

4.7 Relationship Between Alpha Activity and Behavioral Performance

The association between alpha change and recorded behavioral accuracy change from 2-back to 4-back was examined using Spearman's rank correlation. The correlation was negative but not statistically significant ($\rho = -0.3301$, $p = 0.181$). Leave-one-participant-out analyses also produced non-significant correlations. Given the concerns about the validity of the recorded `Matched` labels, this result should be regarded as exploratory and not as a definitive test of the relationship between neural activity and true task performance.

5. Discussion

Among the three frequency bands, alpha activity showed the clearest condition-related modulation, particularly in the 2-back versus 4-back comparison. This finding is compatible with prior work suggesting that alpha activity is sensitive to working-memory demands, while the direction and magnitude of the effect can vary by task and analysis context [3, 5, 6]. The present results do not establish a universal monotonic relationship between alpha power and increasing N-back difficulty: alpha power decreased relative to 1-back at 2-back and 3-back, then increased above the 1-back reference at 4-back.

Theta power did not show a reliable overall condition effect, and the beta effect did not survive the corrected significance threshold. These findings should be interpreted as results from this dataset and analysis pipeline rather than as evidence that theta or beta are not involved in working memory. The spatial alpha analysis identified condition differences across multiple scalp electrodes, but the approximate montage and sensor-level analysis do not support claims about precise cortical sources.

Several limitations should be considered. The sample contained 18 participants. The electrode montage used approximate standard positions rather than individually digitized coordinates. ICA component identification was interpretative because dedicated EOG channels were unavailable, and the ECG sensitivity analysis did not remove cardiac contamination. In addition, a post-hoc audit raised a concern about the publicly available task code's calculation of the `Matched` marker. Because the exact acquisition-time code and the stimulus letters were unavailable for independent reconstruction, behavioral accuracy and the EEG–behavior association cannot be interpreted with the same confidence as the primary EEG spectral results.

6. Conclusion

This study examined EEG oscillatory activity across four N-back conditions. Alpha activity showed the clearest condition-related modulation, particularly between 2-back and 4-back, and the contrast remained significant in leave-one-participant-out analyses. Theta activity showed

no significant condition effect, while the beta effect did not survive correction for multiple comparisons. The alpha change was not significantly associated with the recorded behavioral accuracy change; however, a potential issue in the publicly available task implementation means this behavioral association should be interpreted cautiously. Overall, the strongest evidence in this analysis concerns condition-dependent alpha-band differences at the scalp level, not a simple linear relationship between workload and oscillatory power.

7. References

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